VISHWA PATEL

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EDUCATION

University of Southern California, Los Angeles, CA **Jan 2026 – Present**
Master of Science (M.S.) in Computer Science GPA: 3.66/4.0

- **Grader, Database Systems (CSCI 585):** Design exam questions and grading rubrics; evaluate student submissions.

Gujarat Technological University, Gujarat, India **Aug 2019 – Jun 2023**
Bachelor of Engineering (B.E.) in Computer Engineering GPA: 3.95/4.0

TECHNICAL SKILLS

Languages: Python, Java, C#, C++, C, Go, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks: .NET MVC, Spring Boot, FastAPI, Flask, Node.js, React, Next.js, Angular, GraphQL, REST APIs, Microservices
Data & Systems: PostgreSQL, pgvector, MySQL, MongoDB, Redis, Cassandra, Elasticsearch, RabbitMQ, ETL, Distributed Systems
AI/ML: PyTorch, TensorFlow, scikit-learn, YOLOv8, ONNX, LLMs, RAG, NLP, Prompt Engineering, Agentic AI, MCP
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, CI/CD, Git, Linux, JUnit, Postman, Agile/Scrum
Coursework: Data Structures, Algorithms, OOP, Database Systems, Operating Systems, Machine Learning, Web Technologies

PROFESSIONAL EXPERIENCE

Software Engineer, **Casepoint Pvt. Ltd.**, Gujarat, India **Jul 2023 – Jun 2025**

- Drove a ∼**90% ETL** throughput gain, fixing a **file-locking** bottleneck and releasing **parallel C#/.NET** execution in **CI/CD**.
- Achieved **50–70%** less manual work and **80%** lower latency after creating a **PST repair service** with **retries** and **Redis** staging.
- Delivered **30%** faster reports and **40%+** fewer errors through **legal hold** custom fields and **Oracle, Aspose, VintaSoft** upgrades.
- Collaborated with **5+ teams** on **8+ SDLC features** via **Agile/Scrum** and **code reviews**, earning the **2024 Excellence Award**.
- **Monitored production systems** on-call, resolving **10–20 urgent tickets weekly** with **root-cause debugging** to meet **SLAs**.

Software Engineer Intern, **Casepoint Pvt. Ltd.**, Gujarat, India **Jan 2023 – Jun 2023**

- Developed a **Recruitment System** with **20+ UI screens** in **.NET MVC, C#, PostgreSQL**, and **Kendo UI** with input validation.
- Cut average response time by **35%** and eased high-traffic load by integrating **RabbitMQ** for **asynchronous messaging**.

Software Engineering Intern (Backend), **Amiseq**, Pune, India **Jun 2022 – Jul 2022**

- Launched **10+ RESTful API** endpoints as **Spring Boot** microservices on **PostgreSQL**, handling all **CRUD** operations.
- Caught integration bugs early with **JUnit** unit tests and **Postman** checks for every endpoint.

PROJECTS

GTM Signal Agent | Python, FastAPI, React, PostgreSQL, LLMs | GitHub

- Engineered an **AI agent** turning funding and hiring signals into outreach via **3 LLM backends** (**Groq, Ollama, Anthropic**).
- Added a **risk-scoring engine** that flags stale deals within a rolling **30-day** window and prioritizes follow-ups.

TechHire: AI Job Search Platform | Python, FastAPI, React, PostgreSQL, Groq | GitHub

- Indexed **5,000+** postings at ∼**7ms** per query via a **Python** scraper, **FastAPI** server, and **React** UI with rich filters.
- Tested a **3-model LLM** scorer on **200+ resumes**, showing an **85-point** strong/weak fit spread and rewriting low-rated bullets.

Smart Document Reader | Python, FastAPI, PostgreSQL/pgvector, Ollama | GitHub

- Designed a local-first **RAG** pipeline with **format-aware extraction** and **hybrid dense/lexical retrieval** for PDFs and tables.
- Identified the top setup among **16 configurations** by benchmarking result quality, grounding each answer in a **source citation**.

Sequence Alignment | Python, Dynamic Programming, Divide-and-Conquer | GitHub

- Cut space from **O(mn) to O(m+n)** with identical optimal output using **Needleman-Wunsch DP** and **Hirschberg's** algorithm.
- Profiled **CPU usage** and **peak memory** over varied data sizes, verifying correctness against a **brute-force baseline**.

LEADERSHIP & CAMPUS INVOLVEMENT

- **Orientation Leader, USC Viterbi (VGOL):** Guided **1,200+** new graduate students with **20+** peers, sharing academic resources.
- **Technical Student Assistant, USC ITS (Current):** Troubleshoot hardware, software, and network issues in classrooms and labs.